

Local global hybrid adaptive stabilization via a quadratic Lyapunov function and a transport PDE

Vasilii Gorelov, Nornickel Research

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We consider next system

$$\dot{x} = f(x) + G(x)(u + \Phi(t)\theta^*), \quad x \in \mathbb{R}^n, \quad u \in \mathbb{R}^m, \quad \theta^* \in \mathbb{R}^q, \quad (1)$$

where $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$, $G \in C^1(\mathbb{R}^n, \mathbb{R}^{n \times m})$, $f(0) = 0$, and

$$\Phi \in L^\infty([0, \infty), \mathbb{R}^{m \times q}).$$

is a known regressor. The unknown vector θ^* is assumed to be constant.

Let

$$A := Df(0), \quad B := G(0).$$

Assume that there exists $K \in \mathbb{R}^{m \times n}$ such that

$$A_{\text{cl}} := A + BK$$

is Hurwitz. Choose $P = P^\top > 0$ as the solution of

$$A_{\text{cl}}^\top P + PA_{\text{cl}} = -Q, \quad Q = Q^\top > 0,$$

and define

$$W(x) := x^\top Px. \quad (2)$$

As in the previous case, there exist constants $a_{\text{loc}} > 0$ and $c_{\text{loc}} > 0$ such that

$$\nabla W(x)^\top (f(x) + G(x)Kx) \leq -c_{\text{loc}}W(x), \quad 0 < W(x) \leq a_{\text{loc}}. \quad (3)$$

The number a_{loc} is the state local Lyapunov radius.

Let $\bar{\Gamma} = \bar{\Gamma}^\top > 0$ be fixed and let

$$\Gamma := \gamma \bar{\Gamma}, \quad \gamma > 0.$$

For the parameter estimate $\hat{\theta}(t)$, define

$$\tilde{\theta}(t) := \hat{\theta}(t) - \theta^*$$

and the extended Lyapunov function

$$V_\gamma(x, \tilde{\theta}) := W(x) + \frac{1}{2} \tilde{\theta}^\top \Gamma^{-1} \tilde{\theta} = W(x) + \frac{1}{2\gamma} \tilde{\theta}^\top \bar{\Gamma}^{-1} \tilde{\theta}. \quad (4)$$

Assumption 1. There exists a known compact convex set $\Theta \subset \mathbb{R}^q$ such that

$$\theta^* \in \Theta.$$

The estimate $\hat{\theta}(t)$ is updated by a projection operator

$$\text{Proj}_\Theta(\hat{\theta}, y) := \arg \min_{v \in T_\Theta(\hat{\theta})} \frac{1}{2}(v - y)^\top \Gamma^{-1}(v - y), \text{ where}$$

$$T_\Theta(\hat{\theta}) := \text{cl} \left\{ \lambda(\theta - \hat{\theta}) : \lambda \geq 0, \theta \in \Theta \right\}.$$

which satisfies

$$\hat{\theta}(t) \in \Theta \quad \forall t \geq 0$$

and projection inequality

$$(\hat{\theta} - \theta^\star)^\top \Gamma^{-1} [\text{Proj}_\Theta(\hat{\theta}, y) - y] \leq 0 \quad (5)$$

holds for all $\hat{\theta} \in \Theta$, $\theta^\star \in \Theta$, and all $y \in \mathbb{R}^q$.

Define

$$C_\Theta := \frac{1}{2} \sup_{\hat{\theta}, \theta \in \Theta} (\hat{\theta} - \theta)^\top \bar{\Gamma}^{-1} (\hat{\theta} - \theta). \quad (6)$$

Lemma 1. *Under Assumption 1, along every projected trajectory,*

$$\frac{1}{2} \tilde{\theta}(t)^\top \Gamma^{-1} \tilde{\theta}(t) \leq \frac{C_\Theta}{\gamma}.$$

Proof. Indeed, $\hat{\theta}(t) \in \Theta$ and $\theta^\star \in \Theta$. Hence, by the definition of C_Θ ,

$$\frac{1}{2} (\hat{\theta}(t) - \theta^\star)^\top \bar{\Gamma}^{-1} (\hat{\theta}(t) - \theta^\star) \leq C_\Theta.$$

Since $\Gamma^{-1} = \gamma^{-1} \bar{\Gamma}^{-1}$, this gives

$$\frac{1}{2} \tilde{\theta}(t)^\top \Gamma^{-1} \tilde{\theta}(t) = \frac{1}{2\gamma} \tilde{\theta}(t)^\top \bar{\Gamma}^{-1} \tilde{\theta}(t) \leq \frac{C_\Theta}{\gamma}.$$

□

The local adaptive law is

$$u_{\text{loc}}(x, \hat{\theta}, t) := Kx - \Phi(t)\hat{\theta}, \quad (7)$$

with

$$\dot{\hat{\theta}} = \text{Proj}_\Theta \left(\hat{\theta}, \Gamma \Phi(t)^\top G(x)^\top \nabla W(x) \right). \quad (8)$$

Lemma 2. *Under the local adaptive law (7)–(8),*

$$\dot{V}_\gamma(x, \tilde{\theta}) \leq -c_{\text{loc}} W(x) \quad (9)$$

whenever $0 < W(x) \leq a_{\text{loc}}$. Consequently, for every level $\bar{a} \in (0, a_{\text{loc}}]$, the set

$$\mathcal{E}_{\bar{a}}^\gamma := \left\{ (x, \hat{\theta}) : V_\gamma(x, \hat{\theta} - \theta^\star) \leq \bar{a} \right\} \quad (10)$$

is positively invariant for the local adaptive system, and every trajectory starting in $\mathcal{E}_{\bar{a}}^\gamma$ satisfies

$$x(t) \rightarrow 0 \quad (t \rightarrow +\infty).$$

Proof. Under (7),

$$u_{\text{loc}} + \Phi(t)\theta^\star = Kx - \Phi(t)\hat{\theta} + \Phi(t)\theta^\star = Kx - \Phi(t)\tilde{\theta}.$$

Hence

$$\dot{x} = f(x) + G(x)Kx - G(x)\Phi(t)\tilde{\theta}.$$

Differentiating V_γ , we get

$$\begin{aligned}\dot{V}_\gamma &= \nabla W(x)^\top \dot{x} + \tilde{\theta}^\top \Gamma^{-1} \dot{\tilde{\theta}} \\ &= \nabla W(x)^\top (f(x) + G(x)Kx) - \nabla W(x)^\top G(x)\Phi(t)\tilde{\theta} \\ &\quad + \tilde{\theta}^\top \Gamma^{-1} \text{Proj}_\Theta \left(\hat{\theta}, \Gamma\Phi(t)^\top G(x)^\top \nabla W(x) \right).\end{aligned}$$

Add and subtract $\Gamma\Phi(t)^\top G(x)^\top \nabla W(x)$ inside the projection term. Then

$$\begin{aligned}\dot{V}_\gamma &= \nabla W(x)^\top (f(x) + G(x)Kx) \\ &\quad - \nabla W(x)^\top G(x)\Phi(t)\tilde{\theta} + \tilde{\theta}^\top \Phi(t)^\top G(x)^\top \nabla W(x) \\ &\quad + \tilde{\theta}^\top \Gamma^{-1} \left[\text{Proj}_\Theta \left(\hat{\theta}, \Gamma\Phi(t)^\top G(x)^\top \nabla W(x) \right) - \Gamma\Phi(t)^\top G(x)^\top \nabla W(x) \right].\end{aligned}$$

The second and third terms cancel. By the projection inequality (5), the last term is nonpositive. Therefore

$$\dot{V}_\gamma \leq \nabla W(x)^\top (f(x) + G(x)Kx).$$

Using (3), we obtain

$$\dot{V}_\gamma \leq -c_{\text{loc}} W(x), \quad 0 < W(x) \leq a_{\text{loc}}.$$

Now let $(x(0), \tilde{\theta}(0)) \in \mathcal{E}_a^\gamma$ with $\bar{a} \leq a_{\text{loc}}$. Since

$$W(x) \leq V_\gamma(x, \tilde{\theta}),$$

we have

$$V_\gamma(x(0), \tilde{\theta}(0)) \leq \bar{a} \implies W(x(0)) \leq \bar{a} \leq a_{\text{loc}}.$$

As long as the trajectory remains in $\mathcal{E}_a^\gamma$, one has $W(x(t)) \leq \bar{a} \leq a_{\text{loc}}$, and hence

$$\dot{V}_\gamma \leq -c_{\text{loc}} W \leq 0.$$

Thus V_γ cannot increase past $\bar{a}$, and $\mathcal{E}_a^\gamma$ is positively invariant.

Furthermore,

$$c_{\text{loc}} \int_0^\infty W(x(t)) dt \leq V_\gamma(x(0), \tilde{\theta}(0)).$$

Since $P > 0$, this implies $x \in L^2([0, \infty))$. The state $x(t)$ is bounded because $W(x(t)) \leq \bar{a}$, and $\hat{\theta}(t) \in \Theta$ by projection. Hence $\dot{x}(t)$ is bounded on $[0, \infty)$, so $x(t)$ is uniformly continuous. By Barbalat's lemma,

$$x(t) \rightarrow 0.$$

□

Assumption 2.

$$G(x)^\top \nabla W(x) \neq 0 \quad \text{for all } x \neq 0. \quad (11)$$

Define the outer adaptive law by

$$u_{\text{ext}}(x, \hat{\theta}, t) := -\Phi(t)\hat{\theta} - \frac{W(x) + \nabla W(x)^\top f(x)}{\|G(x)^\top \nabla W(x)\|^2} G(x)^\top \nabla W(x), \quad x \neq 0, \quad (12)$$

with the same projected update

$$\dot{\hat{\theta}} = \text{Proj}_\Theta \left(\hat{\theta}, \Gamma\Phi(t)^\top G(x)^\top \nabla W(x) \right). \quad (13)$$

Lemma 3. Under the outer adaptive law (12)–(13),

$$\dot{V}_\gamma(x, \tilde{\theta}) \leq -W(x) \quad (14)$$

whenever $x \neq 0$.

Proof. Under (12),

$$u_{\text{ext}} + \Phi(t)\theta^* = -\Phi(t)\tilde{\theta} - \frac{W(x) + \nabla W(x)^\top f(x)}{\|G(x)^\top \nabla W(x)\|^2} G(x)^\top \nabla W(x).$$

Therefore,

$$\begin{aligned} \dot{x} &= f(x) - G(x)\Phi(t)\tilde{\theta} \\ &\quad - G(x) \frac{W(x) + \nabla W(x)^\top f(x)}{\|G(x)^\top \nabla W(x)\|^2} G(x)^\top \nabla W(x). \end{aligned}$$

Differentiating V_γ , using the projected update, and adding and subtracting the unprojected update exactly as in the proof of Lemma 2, we obtain

$$\begin{aligned} \dot{V}_\gamma &\leq \nabla W(x)^\top f(x) \\ &\quad - \frac{W(x) + \nabla W(x)^\top f(x)}{\|G(x)^\top \nabla W(x)\|^2} \nabla W(x)^\top G(x) G(x)^\top \nabla W(x). \end{aligned}$$

But

$$\nabla W(x)^\top G(x) G(x)^\top \nabla W(x) = \|G(x)^\top \nabla W(x)\|^2.$$

Thus

$$\dot{V}_\gamma \leq \nabla W(x)^\top f(x) - (W(x) + \nabla W(x)^\top f(x)) = -W(x).$$

□

Lemma 4. Assume Assumption 1 and Assumption 2. Let $\bar{a} > 0$ and choose $\gamma > 0$ such that

$$\frac{C_\Theta}{\gamma} < \bar{a}.$$

Define

$$r_\gamma := \bar{a} - \frac{C_\Theta}{\gamma} > 0. \quad (15)$$

Consider the outer adaptive law (12)–(13) as long as

$$W(x) > r_\gamma.$$

Then the trajectory reaches the implementable switching set

$$\Omega_\gamma := \{x \in \mathbb{R}^n : W(x) \leq r_\gamma\} \quad (16)$$

in finite time. Moreover, at the first entrance time

$$\tau_\gamma := \inf\{t \geq 0 : W(x(t)) \leq r_\gamma\},$$

one has

$$V_\gamma(x(\tau_\gamma), \tilde{\theta}(\tau_\gamma)) \leq \bar{a}.$$

Consequently, switching at time τ_γ to the local adaptive law places the closed-loop trajectory inside

$$\mathcal{E}_a^\gamma = \{(x, \hat{\theta}) : V_\gamma(x, \hat{\theta} - \theta^*) \leq \bar{a}\}.$$

Proof. By the projection property and the definition of C_Θ , along every projected trajectory,

$$\frac{1}{2}\tilde{\theta}(t)^\top \Gamma^{-1}\tilde{\theta}(t) \leq \frac{C_\Theta}{\gamma}.$$

Therefore, for all $t \geq 0$,

$$V_\gamma(x(t), \tilde{\theta}(t)) = W(x(t)) + \frac{1}{2}\tilde{\theta}(t)^\top \Gamma^{-1}\tilde{\theta}(t) \leq W(x(t)) + \frac{C_\Theta}{\gamma}. \quad (17)$$

If

$$W(x(0)) \leq r_\gamma,$$

then $\tau_\gamma = 0$, and by (17),

$$V_\gamma(x(0), \tilde{\theta}(0)) \leq r_\gamma + \frac{C_\Theta}{\gamma} = \bar{a}.$$

Thus the claim is immediate.

Assume now that

$$W(x(0)) > r_\gamma.$$

Then $x(0) \neq 0$, because $r_\gamma > 0$. As long as

$$W(x(t)) > r_\gamma,$$

we also have $x(t) \neq 0$. Hence the outer adaptive law is well defined. By Lemma 3,

$$\dot{V}_\gamma(x(t), \tilde{\theta}(t)) \leq -W(x(t)) < -r_\gamma.$$

We prove finite time entrance by contradiction. Suppose that the trajectory never reaches Ω_γ . Then

$$W(x(t)) > r_\gamma \quad \forall t \geq 0.$$

Consequently,

$$\dot{V}_\gamma(x(t), \tilde{\theta}(t)) \leq -r_\gamma \quad \forall t \geq 0.$$

Integrating, we get

$$V_\gamma(x(t), \tilde{\theta}(t)) \leq V_\gamma(x(0), \tilde{\theta}(0)) - r_\gamma t.$$

The right-hand side becomes negative for sufficiently large t , whereas

$$V_\gamma(x(t), \tilde{\theta}(t)) \geq 0$$

by definition. This contradiction proves that $\tau_\gamma < \infty$.

At $t = \tau_\gamma$, by definition of the first entrance time,

$$W(x(\tau_\gamma)) \leq r_\gamma.$$

Using again the parameter bound,

$$\begin{aligned} V_\gamma(x(\tau_\gamma), \tilde{\theta}(\tau_\gamma)) &= W(x(\tau_\gamma)) + \frac{1}{2}\tilde{\theta}(\tau_\gamma)^\top \Gamma^{-1}\tilde{\theta}(\tau_\gamma) \\ &\leq r_\gamma + \frac{C_\Theta}{\gamma} \\ &= \bar{a}. \end{aligned}$$

Therefore

$$(x(\tau_\gamma), \hat{\theta}(\tau_\gamma)) \in \mathcal{E}_\bar{a}^\gamma.$$

□

Theorem 1. *Assume:*

1. $A + BK$ is Hurwitz;
2. local estimate (3) holds on $0 < W \leq a_{\text{loc}}$;
3. Assumption 1 holds;
4. Assumption 2 holds.

Then for every desired Lyapunov margin

$$0 < \eta < a_{\text{loc}},$$

there exist a local Lyapunov level $\bar{a}$, an implementable switching level r_γ , and a gain threshold $\gamma_\eta > 0$ such that for every $\gamma \geq \gamma_\eta$, the following hybrid adaptive controller is well defined:

$$\text{use } u_{\text{ext}} \text{ while } W(x) > r_\gamma, \quad \text{switch to } u_{\text{loc}} \text{ once } W(x) \leq r_\gamma.$$

The outer adaptive trajectory reaches the switching set

$$\Omega_\gamma = \{x : W(x) \leq r_\gamma\}$$

in finite time. After the switch, the trajectory belongs to

$$\mathcal{E}_a^\gamma = \{(x, \hat{\theta}) : V_\gamma(x, \hat{\theta} - \theta^*) \leq \bar{a}\},$$

remains in this set for all future time under the local adaptive law, and satisfies

$$x(t) \rightarrow 0 \quad (t \rightarrow +\infty).$$

More precisely, one may choose

$$\bar{a} := a_{\text{loc}} - \eta, \tag{18}$$

and any

$$\gamma_\eta > \frac{C_\Theta}{\bar{a}}. \tag{19}$$

For every $\gamma \geq \gamma_\eta$, define

$$r_\gamma := \bar{a} - \frac{C_\Theta}{\gamma} > 0. \tag{20}$$

Proof. Fix $\eta \in (0, a_{\text{loc}})$, and define

$$\bar{a} := a_{\text{loc}} - \eta.$$

Then

$$0 < \bar{a} < a_{\text{loc}}.$$

Hence

$$V_\gamma(x, \tilde{\theta}) \leq \bar{a} \implies W(x) \leq \bar{a} < a_{\text{loc}}.$$

Thus, once the trajectory reaches $\mathcal{E}_a^\gamma$, the state lies strictly inside the local Lyapunov region where Lemma 2 applies.

Choose γ_η so that

$$\gamma_\eta > \frac{C_\Theta}{\bar{a}}.$$

Then for every $\gamma \geq \gamma_\eta$,

$$\frac{C_\Theta}{\gamma} < \bar{a}.$$

Therefore the number

$$r_\gamma := \bar{a} - \frac{C_\Theta}{\gamma}$$

is strictly positive.

We use the following hybrid rule. If

$$W(x(0)) \leq r_\gamma,$$

then the local adaptive law is used immediately. In this case, by the parameter bound,

$$\begin{aligned} V_\gamma(x(0), \tilde{\theta}(0)) &= W(x(0)) + \frac{1}{2} \tilde{\theta}(0)^\top \Gamma^{-1} \tilde{\theta}(0) \\ &\leq r_\gamma + \frac{C_\Theta}{\gamma} \\ &= \bar{a}. \end{aligned}$$

Thus

$$(x(0), \hat{\theta}(0)) \in \mathcal{E}_a^\gamma.$$

Lemma 2 then implies positive invariance of $\mathcal{E}_a^\gamma$ and convergence $x(t) \rightarrow 0$.

Now suppose that

$$W(x(0)) > r_\gamma.$$

Then the outer adaptive law is used as long as

$$W(x) > r_\gamma.$$

By Lemma 4, the trajectory reaches

$$\Omega_\gamma = \{x : W(x) \leq r_\gamma\}$$

in finite time. Let

$$\tau_\gamma := \inf\{t \geq 0 : W(x(t)) \leq r_\gamma\}$$

be the first entrance time. Lemma 4 gives

$$V_\gamma(x(\tau_\gamma), \tilde{\theta}(\tau_\gamma)) \leq \bar{a}.$$

Therefore

$$(x(\tau_\gamma), \hat{\theta}(\tau_\gamma)) \in \mathcal{E}_a^\gamma.$$

At time τ_γ , switch to the local adaptive law. Since

$$\bar{a} < a_{\text{loc}},$$

Lemma 2 implies that $\mathcal{E}_a^\gamma$ is positively invariant under the local adaptive law. Consequently,

$$V_\gamma(x(t), \tilde{\theta}(t)) \leq \bar{a} \quad \forall t \geq \tau_\gamma,$$

and therefore

$$W(x(t)) \leq \bar{a} < a_{\text{loc}} \quad \forall t \geq \tau_\gamma.$$

Again by Lemma 2,

$$x(t) \rightarrow 0 \quad (t \rightarrow +\infty).$$

□

Remark 1. *The theorem shows the precise role of the adaptive gain. The compact parameter set produces the parameter bound*

$$\frac{1}{2}\tilde{\theta}^\top \Gamma^{-1}\tilde{\theta} \leq \frac{C_\Theta}{\gamma}.$$

By increasing γ , this bound can be made smaller than any prescribed local level $\bar{a} > 0$. The implementable switching level is then chosen as

$$r_\gamma = \bar{a} - \frac{C_\Theta}{\gamma} > 0.$$

Thus

$$W(x) \leq r_\gamma \implies V_\gamma(x, \tilde{\theta}) \leq \bar{a},$$

even though V_γ itself depends on the unknown parameter θ^ . The controller therefore switches using only the measurable quantity $W(x)$, while the proof uses the extended Lyapunov function to certify entrance into the local invariant set.*

1 Example: adaptive roll stabilization of a ship

We now illustrate the proposed hybrid adaptive design on a nonlinear roll model of a ship. The state is

$$x = \begin{bmatrix} \theta \\ p \end{bmatrix} \in \mathbb{R}^2,$$

where θ is the roll angle and $p = \dot{\theta}$ is the roll rate. The control input is a scalar torque $u \in \mathbb{R}$. The dynamics are

$$\dot{\theta} = p, \quad \dot{p} = -cp - k \sin \theta + u + d(t), \quad (21)$$

where $c > 0$ is the damping coefficient, $k > 0$ is the restoring coefficient, and $d(t)$ is an unknown matched wave induced disturbance.

Equivalently,

$$\dot{x} = f(x) + B(u + d(t)), \quad (22)$$

where

$$f(x) = \begin{bmatrix} p \\ -cp - k \sin \theta \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The parameters are chosen as

$$c = 0.85, \quad k = 2.20.$$

To describe the wave excitation, let y denote the transverse spatial coordinate across the ship beam, measured relative to the centerline. The constant $b > 0$ is the ship beam, so the points $y = \pm b/2$ correspond to the right and left sides of the hull. The scalar $\eta(y, t)$ denotes the wave surface elevation at position y and time t . The coefficients $A_i > 0$ are harmonic amplitudes, $\omega_i > 0$ are temporal frequencies, $k_i > 0$ are wave numbers, and $\psi_i \in [0, 2\pi)$ are phase shifts. The constant $\mu_{\text{wave}} > 0$ converts the left-right wave elevation difference into a matched roll moment acting on the ship.

In the numerical example we take

$$b = 2.8, \quad \mu_{\text{wave}} = 1.0, \quad N = 30.$$

The frequencies are uniformly distributed over the interval

$$\omega_i \in [0.35, 3.50], \quad i = 1, \dots, N,$$

and the wave numbers are chosen according to

$$k_i = 0.65 \omega_i.$$

The wave profile is modeled as a finite harmonic superposition

$$\eta(y, t) = \sum_{i=1}^N A_i \sin(k_i y - \omega_i t + \psi_i). \quad (23)$$

The corresponding normalized roll moment is defined as the difference between the wave elevations at the right and left sides of the hull:

$$d(t) = \mu_{\text{wave}} \left(\eta\left(\frac{b}{2}, t\right) - \eta\left(-\frac{b}{2}, t\right) \right). \quad (24)$$

Using the trigonometric identity

$$\sin(\alpha + \beta) - \sin(-\alpha + \beta) = 2 \sin(\alpha) \cos(\beta),$$

we obtain

$$d(t) = \sum_{i=1}^N 2\mu_{\text{wave}} A_i \sin\left(\frac{k_i b}{2}\right) \cos(\omega_i t - \psi_i). \quad (25)$$

To match the adaptive-control model, we expand each harmonic in the paired sine-cosine basis

$$\{\sin(\omega_i t), \cos(\omega_i t)\}_{i=1}^N,$$

that is, in a finite-dimensional Fourier basis. Using

$$\cos(\omega_i t - \psi_i) = \sin(\psi_i) \sin(\omega_i t) + \cos(\psi_i) \cos(\omega_i t),$$

the disturbance can be written in the matched parametric form

$$d(t) = \varphi(t)^\top \theta^*, \quad (26)$$

where $\theta^* \in \mathbb{R}^{2N}$ is an unknown constant parameter vector and the regressor is

$$\varphi(t) = \begin{bmatrix} \sin(\omega_1 t) \\ \cos(\omega_1 t) \\ \vdots \\ \sin(\omega_N t) \\ \cos(\omega_N t) \end{bmatrix} \in \mathbb{R}^{2N}. \quad (27)$$

Thus, in this example, the parameter dimension is

$$q = 2N = 60.$$

The unknown vector θ^* collects the coefficients of this Fourier representation. Its entries are given by

$$\theta_{2i-1}^* = 2\mu_{\text{wave}} A_i \sin\left(\frac{k_i b}{2}\right) \sin(\psi_i), \quad \theta_{2i}^* = 2\mu_{\text{wave}} A_i \sin\left(\frac{k_i b}{2}\right) \cos(\psi_i), \quad (28)$$

for $i = 1, \dots, N$. Hence the adaptive controller does not estimate the wave profile $\eta(y, t)$ directly; instead, it estimates the coefficients of the disturbance in the chosen finite Fourier basis.

The compact parameter set is chosen as the Euclidean ball

$$\Theta = \{\theta \in \mathbb{R}^q : \|\theta\| \leq R_\Theta\}, \quad R_\Theta = 0.65. \quad (29)$$

We take

$$\bar{\Gamma} = I_q, \quad \Gamma = \gamma I_q.$$

Since $\hat{\theta}(t) \in \Theta$ and $\theta^* \in \Theta$, one has

$$\|\hat{\theta}(t) - \theta^*\| \leq 2R_\Theta.$$

Hence

$$C_\Theta = 2R_\Theta^2. \quad (30)$$

For $R_\Theta = 0.65$, this gives

$$C_\Theta = 2(0.65)^2 = 0.845.$$

The linearization of (22) at the origin is

$$A = \begin{bmatrix} 0 & 1 \\ -k & -c \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

For the chosen numerical parameters,

$$A = \begin{bmatrix} 0 & 1 \\ -2.20 & -0.85 \end{bmatrix}.$$

We choose the local feedback

$$K = \begin{bmatrix} -3.8 & -2.4 \end{bmatrix}. \quad (31)$$

Then

$$A_{\text{cl}} = A + BK = \begin{bmatrix} 0 & 1 \\ -6.0 & -3.25 \end{bmatrix}.$$

Its eigenvalues are

$$\lambda_{1,2} = -1.625 \pm 1.8329i.$$

Therefore A_{cl} is Hurwitz.

Let $P = P^\top > 0$ be the solution of the Lyapunov equation

$$A_{\text{cl}}^\top P + PA_{\text{cl}} = -I. \quad (32)$$

Numerically,

$$P = \begin{bmatrix} 1.34775641 & 0.08333333 \\ 0.08333333 & 0.17948718 \end{bmatrix}. \quad (33)$$

The state Lyapunov function is

$$W(x) = x^\top Px. \quad (34)$$

Since A_{cl} is Hurwitz there exists $a_{\text{loc}} > 0$ such that the nominal local decay estimate

$$\nabla W(x)^\top (f(x) + BKx) \leq -c_{\text{loc}} W(x)$$

holds for $0 < W(x) \leq a_{\text{loc}}$. In the example we take

$$a_{\text{loc}} = 0.20.$$

The local adaptive controller is

$$u_{\text{loc}}(x, \hat{\theta}, t) = Kx - \varphi(t)^\top \hat{\theta}. \quad (35)$$

The projected adaptive law is

$$\dot{\hat{\theta}} = \text{Proj}_\Theta \left(\hat{\theta}, \Gamma \varphi(t) B^\top \nabla W(x) \right). \quad (36)$$

Since

$$\nabla W(x) = 2Px,$$

we have

$$B^\top \nabla W(x) = 2B^\top Px.$$

Thus, in explicit form,

$$\dot{\hat{\theta}} = \text{Proj}_\Theta \left(\hat{\theta}, 2\gamma \varphi(t) B^\top Px \right). \quad (37)$$

The corresponding extended Lyapunov function is

$$V_\gamma(x, \tilde{\theta}) = x^\top Px + \frac{1}{2\gamma} \|\tilde{\theta}\|^2, \quad \tilde{\theta} = \hat{\theta} - \theta^*. \quad (38)$$

By the general local adaptive result, for all $0 < W(x) \leq a_{\text{loc}}$,

$$\dot{V}_\gamma(x, \tilde{\theta}) \leq -c_{\text{loc}} W(x).$$

Therefore every sublevel set

$$\mathcal{E}_a^\gamma = \left\{ (x, \hat{\theta}) : V_\gamma(x, \hat{\theta} - \theta^*) \leq \bar{a} \right\}, \quad 0 < \bar{a} < a_{\text{loc}},$$

is positively invariant under the local adaptive closed-loop system, and

$$x(t) \rightarrow 0 \quad (t \rightarrow +\infty).$$

In the outer region, the adaptive controller is chosen according to the transport construction. Since the input is scalar, define

$$g(x) := B^\top \nabla W(x) = 2B^\top Px. \quad (39)$$

The theoretical outer law is

$$u_{\text{ext}}(x, \hat{\theta}, t) = -\varphi(t)^\top \hat{\theta} - \frac{W(x) + \nabla W(x)^\top f(x)}{g(x)}, \quad (40)$$

whenever $g(x) \neq 0$. Equivalently, this is the scalar-input specialization of

$$u_{\text{ext}} = -\varphi(t)^\top \hat{\theta} - \frac{W(x) + \nabla W(x)^\top f(x)}{\|B^\top \nabla W(x)\|^2} B^\top \nabla W(x).$$

The adaptive update in the outer mode is

$$\dot{\hat{\theta}} = \text{Proj}_\Theta \left(\hat{\theta}, \Gamma \varphi(t) B^\top \nabla W(x) \right). \quad (41)$$

Thus the same projected estimator is used in both modes.

Substituting (40) into the dynamics gives

$$\dot{x} = f(x) - B\varphi(t)^\top \tilde{\theta} - B \frac{W(x) + \nabla W(x)^\top f(x)}{B^\top \nabla W(x)}.$$

Using the adaptive update and the projection inequality, one obtains

$$\dot{V}_\gamma(x, \tilde{\theta}) \leq \nabla W(x)^\top f(x) - (W(x) + \nabla W(x)^\top f(x)) = -W(x),$$

provided $B^\top \nabla W(x) \neq 0$. Hence, as long as the outer law is well defined,

$$\dot{V}_\gamma(x, \tilde{\theta}) \leq -W(x). \quad (42)$$

For the present single-input two-dimensional example,

$$B^\top \nabla W(x) = 2 \begin{bmatrix} 0 & 1 \end{bmatrix} P \begin{bmatrix} \theta \\ p \end{bmatrix} = 2(P_{21}\theta + P_{22}p).$$

Therefore the nondegeneracy condition is not global in $\mathbb{R}^2$, since $B^\top \nabla W(x)$ may vanish on a line through the origin. The rigorous application of the theorem is therefore understood on any outer region, or along any outer trajectory, where

$$|B^\top \nabla W(x)| > 0.$$

We choose

$$\eta = 0.06, \quad a_{\text{loc}} = 0.20.$$

Therefore

$$\bar{a} = a_{\text{loc}} - \eta = 0.14. \quad (43)$$

Since

$$C_\Theta = 0.845,$$

the gain threshold is chosen as

$$\gamma_\eta > \frac{C_\Theta}{\bar{a}} = \frac{0.845}{0.14} \approx 6.0357. \quad (44)$$

In the simulation,

$$\gamma_\eta = 6.3375, \quad \gamma = 7.921875.$$

Hence

$$\frac{C_\Theta}{\gamma} = \frac{0.845}{7.921875} = 0.106666 \dots < \bar{a}.$$

The implementable switching level is therefore

$$r_\gamma = \bar{a} - \frac{C_\Theta}{\gamma} = 0.14 - 0.106666 \dots = 0.033333 \dots \quad (45)$$

The hybrid adaptive controller is

$$u(t) = \begin{cases} u_{\text{ext}}(x(t), \hat{\theta}(t), t), & W(x(t)) > r_\gamma, \\ u_{\text{loc}}(x(t), \hat{\theta}(t), t), & W(x(t)) \leq r_\gamma. \end{cases} \quad (46)$$

The switch is performed only once. After the first entrance into the set

$$\Omega_\gamma = \{x \in \mathbb{R}^2 : W(x) \leq r_\gamma\},$$

the controller remains in the local adaptive mode.